Revenue Analysis for Solstice Opal Hotel

Objective

Identify which guest segments generate the highest ancillary revenue and recommend actions to maximise future revenue.

Data Validation

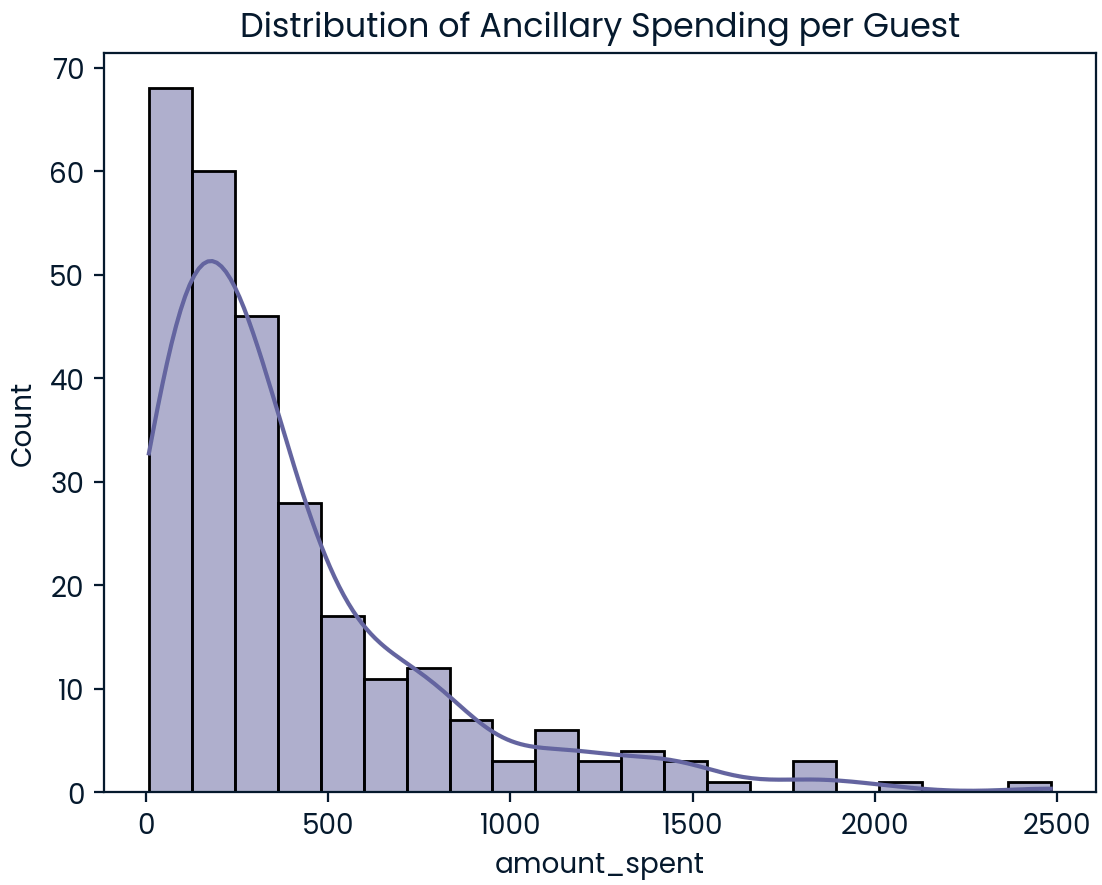
- Data validation was performed to ensure consistency and reliability across the three source systems (guest profiles, stay details, and ancillary spend).
- The guest_id format in ancillary_spend was standardised using regex extraction to align with the other source tables.
- Missing values across all datasets were identified and standardised to NaN to ensure consistent handling during analysis.
- Categorical variables, including loyalty_tier, booking_channel, and reason_for_stay, were validated against expected domain definitions, with no unexpected categories observed, confirming data consistency across systems.
- The check_in_date field was converted into datetime format to ensure temporal consistency for time-based analysis.
- A sanity check confirmed that amount_spent contained no negative values, ensuring the integrity of revenue data.
- Due to differences in granularity between transaction-level ancillary spend and stay-level data, the analysis was aggregated at guest level to create a consistent basis for behavioural segmentation. Guest-level aggregation was used to compare spending patterns across different customer segments.

Business Metric

Average Revenue Per Guest (ARPG)

- Definition Average ancillary revenue generated by each guest.
- Formula Total ancillary revenue / Number of guests

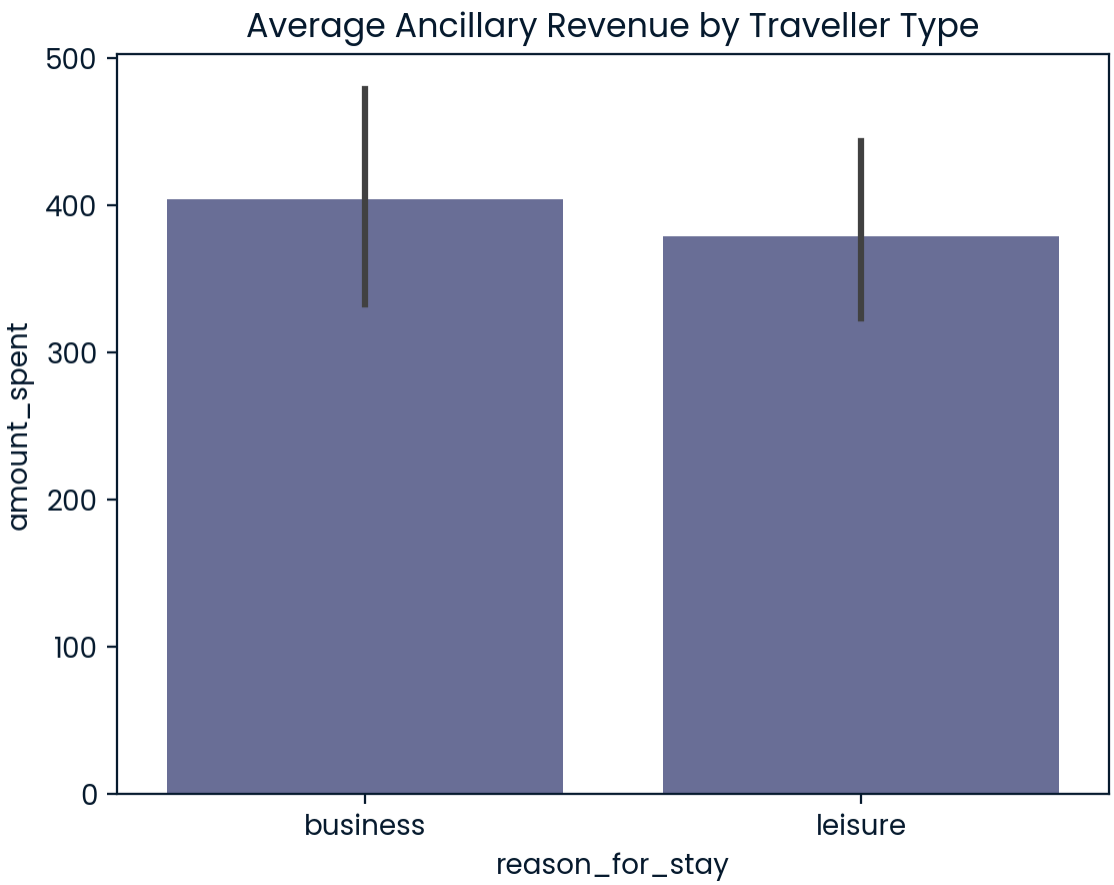
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Current Average Ancillary Revenue per Guest (ARPG) : $390.22
Most guests spend relatively small amounts while a few guests spend considerably
more, indicating a right-skewed distribution.
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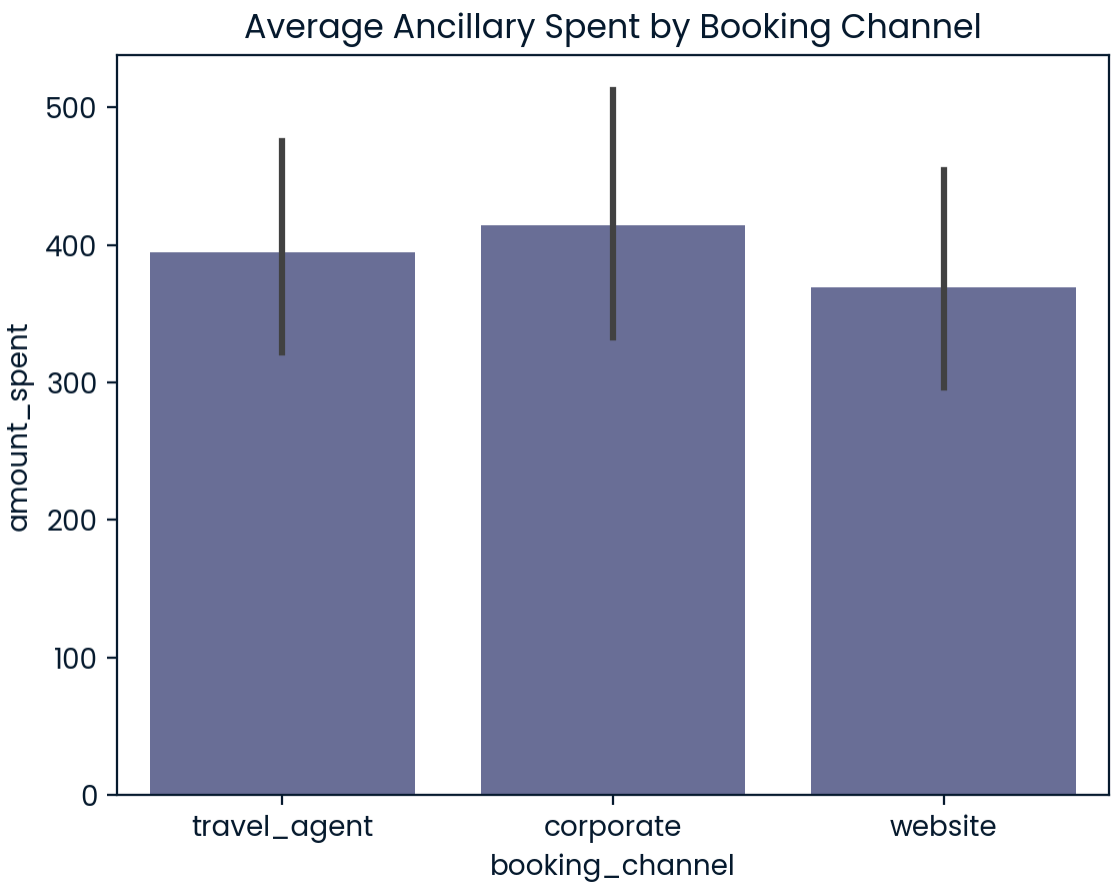
High-Value Guest Analysis

Guest Segment Analysis

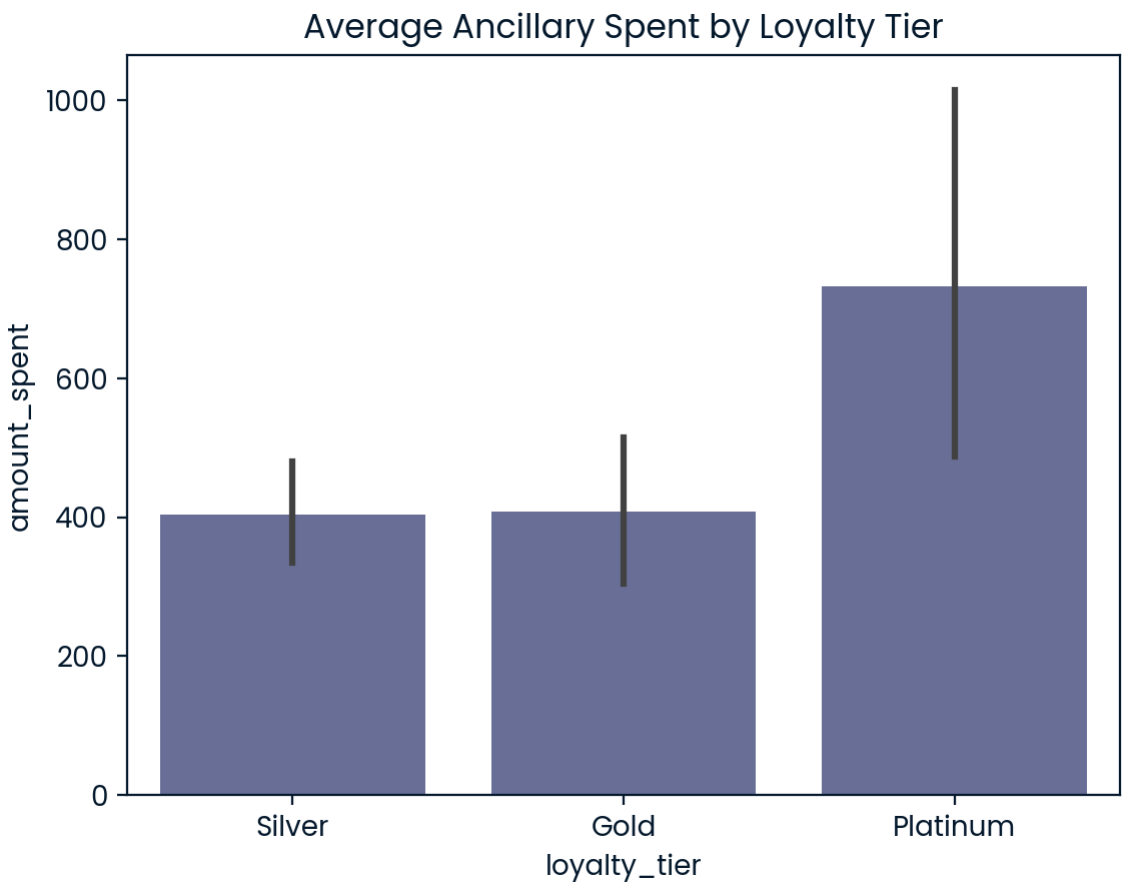
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Average Ancillary Revenue by Traveller Type:
Business: $404.14
Leisure: $378.88
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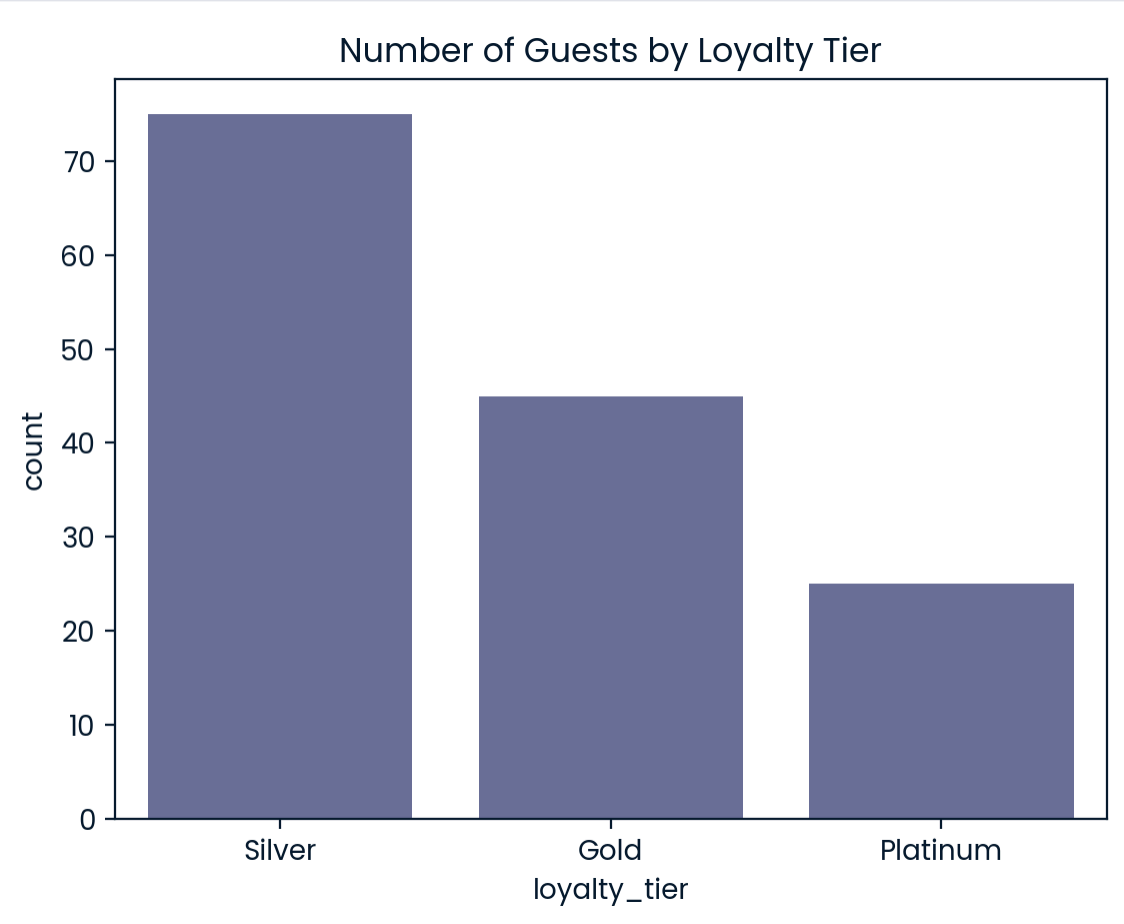



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Average Ancillary Spent by Booking Channel:
Corporate: $414.14
Travel_Agent: $394.86
Website: $369.23
Corporate guests spend 12.2% more than Website guests
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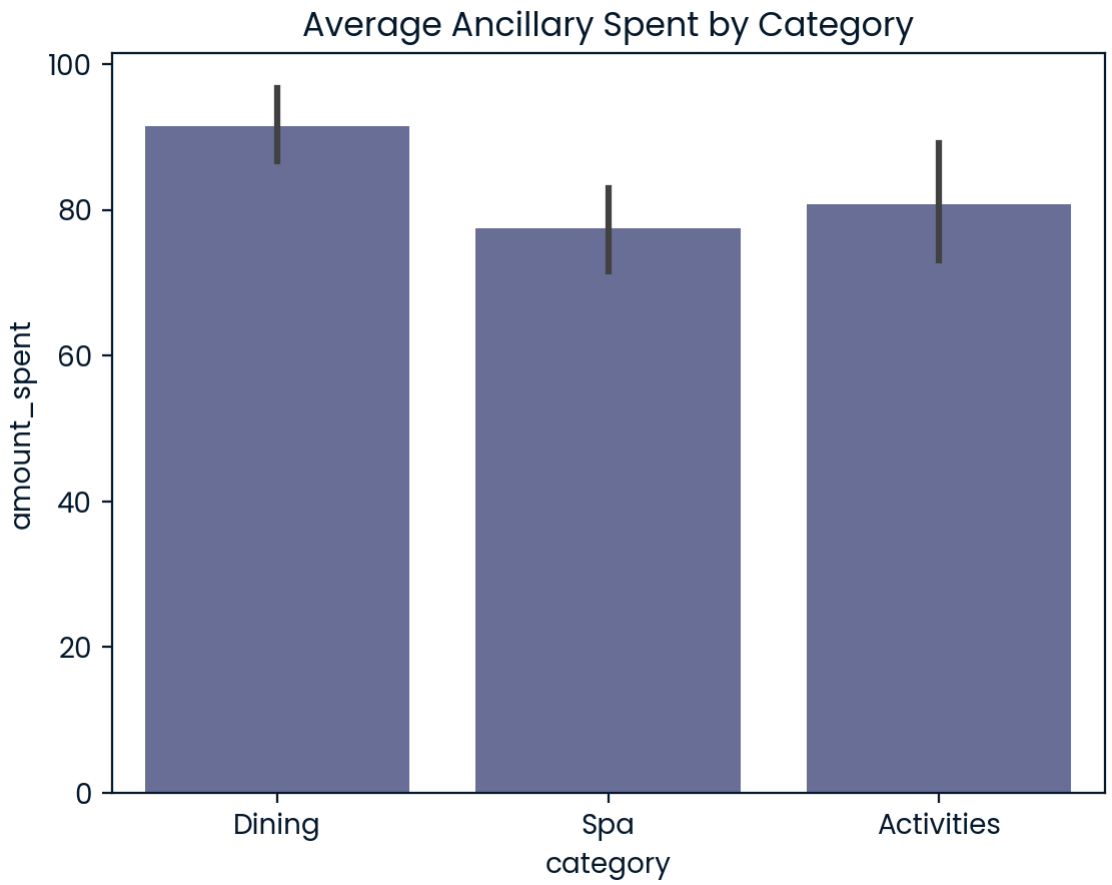

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Average Ancillary Spent by Loyalty Tier:
Gold: $407.62
Platinum: $731.86
Silver: $404.09
Platinum members spent  81.1% more than Silver members
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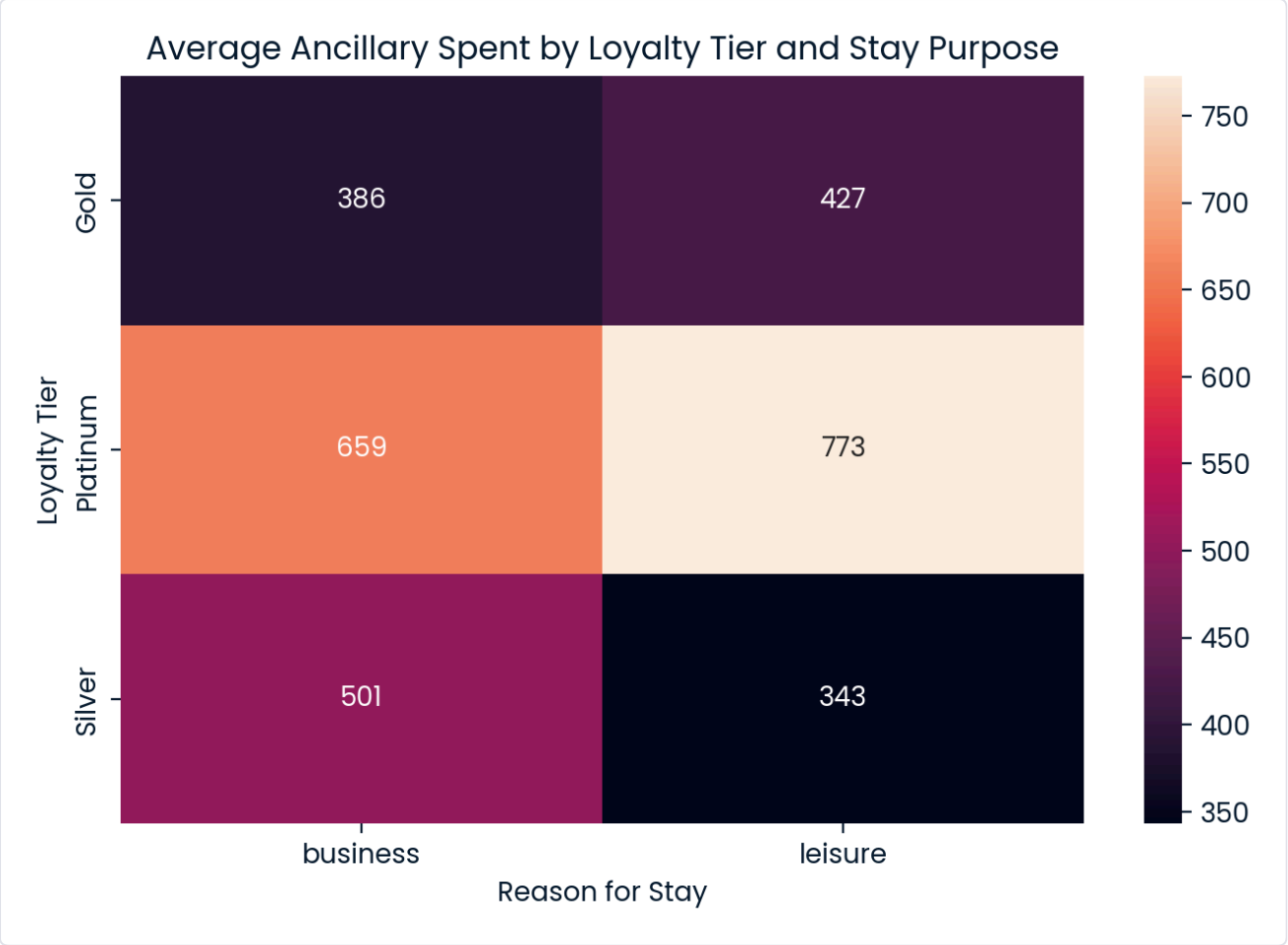


Ancillary Service Analysis

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Average Ancillary Spent by Category:
Activities: $382.64
Dining: $422.84
Spa: $298.35
```



Segment Interaction Analysis



Key Findings

- Current Average Ancillary Revenue per Guest (ARPG) is $390.22
- Business travellers generate higher average ancillary spend than leisure travellers, indicating stronger upsell potential.
- Platinum members have the highest average ancillary spend, spending 81.1% more than Silver members.
- Guests booking through corporate channels generate 12.2% higher average ancillary spend compared with website bookings.
- Spending is highly concentrated among a small number of guests.
- The top 10% highest-spending guests contribute 35.31% of total ancillary revenue, showing revenue concentration among high-value guests.

Recommendation 1

Target Platinum business travellers before arrival.

Because: Platinum highest ARPG & Business travellers highest spending

Action:

- Create pre-arrival upsell campaigns targeting Platinum business travellers, offering dining credits, spa packages, and activity bundles before check-in.

Recommendation 2

Optimise corporate booking channels with embedded ancillary packages.

Because: Corporate channel guests have 12.2% higher average ancillary spend than website bookings.

Action:

- Add optional dining and activity bundles during corporate booking process
- Promote packages through corporate booking partners
- Include add-ons as part of business travel arrangements

Recommendation 3

Create a VIP retention programme targeting the top 10% highest-spending guests with personalised offers and loyalty benefits.

Because: Top 10% guests generate 35% revenue.

Action:

- Exclusive dining experiences
- Spa packages
- Early access offers
- Personalised recommendations

Conclusion

A small proportion of guests (top 10%) contribute approximately 35.31% of total ancillary revenue, indicating that revenue is concentrated among high-value customers.

To maximise ancillary revenue, Solstice Opal Hotel should focus on high-value guest segments, particularly Platinum members, business travellers, and corporate-channel guests. By combining targeted upselling strategies with VIP retention initiatives, the hotel can increase ancillary revenue while strengthening relationships with its most valuable customers.